

Prerana Kottapalli

PRAY-ruh-naa COTH-uh-PUH-li | prerana0207@gmail.com

 [pk0207.github.io](https://github.com/pk0207) |  [prerana-kottapalli](https://www.linkedin.com/in/prerana-kottapalli) |  [PK0207](https://github.com/PK0207) |  [Prerana Kottapalli](https://orcid.org/Prerana-Kottapalli)

EDUCATION

- **University of Florida** January 2025 - June 2030
PhD in Astronomy Gainesville, FL, USA
- **Durham University** September 2024
MSc in Scientific Computing and Data Analysis in Astrophysics Durham, United Kingdom
 - Grade: Distinction
- **University of California Santa Barbara** June 2022
BSc in Physics with a Minor in Astronomy Santa Barbara, CA, USA

PUBLICATIONS

- [1] *Poster* Prerana Kottapalli, et al. (2023). **Investigation and Characterization of Random Telegraph Signal on CMOS Detectors on New LCO Telescope Installations for the BANZAI Data Reduction Pipeline.** In AAS meeting 241, January 2023, Seattle.
- [2] Alex Thomas, Natalie LeBaron + **Prerana Kottapalli** et al. (2025). **The Local Galactic Transient Survey Applied to an Optical Search for Directed Intelligence.**
- [3] *Technical Document* Alex Thomas, Natalie LeBaron + **Prerana Kottapalli** et al. (2025). **TRIPP: A General Purpose Data Pipeline for Astronomical Image Processing.** Supporting technical document for submitted paper.
- [4] *Python Code Package* Prerana Kottapalli (March 3, 2023). **cmos-noise-map.** A tool to create a read noise map for CMOS detectors by modelling Random Telegraph Signal. This was originally created for use by LCOGT's BANZAI pipeline. [Read the docs](#)

RESEARCH EXPERIENCE

- **Graduate Research Assistant** October 2024 - Present
Dr. Keri Hoadley, UV Space Laboratory University of Florida, Gainesville, USA
 - Benchmarking data quality requirements for Hubble Space Telescope Ultraviolet (HST UV) observations of small-scale gas dynamics.
 - Analyzing HST UV data to investigate molecular gas dynamics around young stars at the data resolution limit (15 km/s).
 - Developing a protoplanetary disk radiative transfer model to incorporate disk winds.
- **Master's Dissertation Research** December 2023 - September 2024
Dr. Azadeh Fattahi-Savadjani, Theoretical Cosmology Durham University, UK
 - Discovered significant domain decomposition issues slowing down performance by comparing the performance of the cosmological N-body hydrodynamic codes.
 - Simulated dwarf galaxy formation using a semi-analytical model GALFORM.
 - Analyzed dwarf galaxy populations in simulations and found link between environment overdensity and distance of satellite galaxy from its host.
- **Science Operations and Data Analyst** March 2022 - March 2023
Las Cumbres Observatory  Goleta, CA, USA
 - Developed echelle spectrograph data reduction, achieving 40% improvement in precision.
 - Used machine learning to model and predict CMOS imaging sensor noise, improving uncertainty quantification in image data products.
 - Packaged the new feature: cmos-noise-map toolkit in Python to quickly generate visualizations of data to explore quality of spectra, and writing detailed documentation.
 - Integrated package into existing data reduction pipeline.
- **Undergraduate Research Assistant** January 2019 - January 2023
Dr. Philip Lubin, Deepspace Cosmology Goleta, CA, USA
 - Developed a difference image analysis based real time fast-transient searching pipeline. 6 Megabyte images are processed in < 2s, with a 20% 5 σ detection rate.
 - Developed a multichroic camera that simultaneously captures images in SDSS g', r', and i' bands.
 - Constructed periodicity detection technique using a Fast-Fourier Transform (FFT) algorithm.

TALKS AND POSTER PRESENTATIONS

- **Talk: The Problem with Pluto** 
Astronomy on Tap Gainesville, September 2025
- **Poster: Investigation and Characterization of Random Telegraph Signal on CMOS Detectors on New LCO Telescope Installations for the BANZAI Data Reduction Pipeline** 
AAS 241, Seattle, January 2023
- **Talk: 1,000 Pictures, 10,000 Pictures** 
Astronomy on Tap Santa Barbara, February 2023
- **Talk: Lightcurve Construction Techniques for Intelligent Light Sources** 
Kavli Institute for Theoretical Physics, September 2021

TEACHING AND OUTREACH

- **President of the Graduate Astronomy Organization** *August 2025 - Present*
University of Florida
- **Graduate Teaching Assistant** *January 2025 - Present*
University of Florida
- **Course DEI representative for the Computer Science Department** *September 2023 - June 2024*
University of Durham
- **Learning Assistant** *June 2021 - March 2022*
University of California Santa Barbara
- **Vice-President and President of Undergraduate Diversity and Inclusion and Physics** *June 2020 - June 2022*
University of California Santa Barbara
- **Co-founder Astronomy Society at UCSB** *September 2019 - June 2022*
University of California Santa Barbara

SKILLS

- **Programming Languages:** Python, R, C, C++, UNIX, HTML5, CSS, JavaScript, OpenMP, MPI, CUDA, SYCL, SQL
- **Web Technologies:** GitHub, Google Cloud Platform, MS Office Applications
- **Data Science & Machine Learning:** Convolutional Neural Networks (CNN), Generative Adversarial Networks (GAN), Gaussian Process Regression, tensorflow, pytorch
- **Specialized Area:** IRAF, SExtractor, SAOImageS, AstroImageJ
- **Mathematical & Statistical Tools:** Mathematica, Wolfram Alpha

HONORS AND AWARDS

- **Central Coast Data Science Fellowship** *2021*
University of California Santa Barbara 
 - A program to promote the application of data science to real-life projects
 - Developed a [web application](#) (CalCOFI.io: Capstone App) for the California Cooperative Oceanic Fisheries Investigations.
- **Chairman's Appreciation Award** *2022*
Department of Physics, UCSB
 - An award for outstanding contributions that enhance the UCSB Physics Community
- **The UCSB Edison STEM Research Scholarship** *2020*
University of California Santa Barbara
 - Research grant to fund the development and installation of the multichroic camera
- **Dean's Honors for High Achiever** *2020*
University of California Santa Barbara